SafeNav

Adlin Hamdan, Fatma Azzab, Nithila Balaji, Yuhan Wang, Zhenya Ratushko

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Background

What inspired SafeNav?

Google Maps is great for getting you from Point A to Point B, but have you ever felt unsafe on your journey or thought: Surely, there's gotta be a better way - why am I on this dim lit road? Introducing SafeNav: a dynamic map takes into consideration the weather, past crime occurrences, and other factors on the streets to get you to where you need to go safely and efficiently.

02

Build process

Collecting and wrangling data

- Extracting real-time weather data for Madison, WI using an API call to weather.gov
    - Calculating weather factors -> determining best routes
- Building a web-scraping function to scrape data from the City of Madison police report database (notable incidents from 01/01/24 - 01/01/25)
    - Grouping by low/medium/ high severity and time of day -> determining best routes

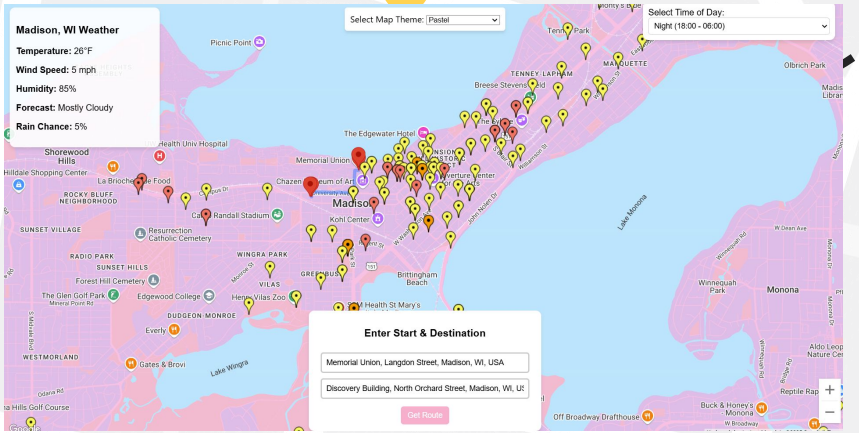
Released	Incident	Case Number	Address	Released By	Time of Incident
02/19/2024	Weapons Violation	2024-00067954	7500 block Westward Way	P.O. Korrie Rondorf	20:58:00
02/20/2024	Robbery	2024-00071935	2900 block Coho St	P.O. Korrie Rondorf	20:02:00
02/22/2024	Arrested Person	2024-00073417	500 block State St	P.O. Korrie Rondorf	19:05:00
02/22/2024	Arrested Person	2024-00075091	2500 block Fich Hatchery Rd	P.O. Korrie Rondorf	20:37:00
02/22/2024	Weapons Violation	2024-00072439	1400 block McKenna Blvd	P.O. Korrie Rondorf	08:23:00
02/22/2024	Robbery	2024-00074134	5700 block Raymond Rd	P.O. Korrie Rondorf	09:27:00
02/22/2024	Disturbance	2024-00075412	100 block Dayton St	P.O. Korrie Rondorf	02:32:00
02/22/2024	Disturbance	2024-00075412	100 block Dayton St	P.O. Korrie Rondorf	02:32:00
02/23/2024	Missing Adult	2024-076800	1600 Northport Dr	P.O. Anthony Vogel	20:13:00
02/23/2024	Sexual Assault	2005-124598	2000 Block Eastwood Dr.	P.O. Anthony Vogel	20:13:00
02/23/2024	Disturbance	2024-076428	600 E Washington Ave	P.O. Anthony Vogel	16:10:00
02/27/2024	Disturbance	2024-81200	400 block W. Gilman St.	PIO Stephanie Fryer	20:20:00
03/01/2024	Battery	2024-87495	1800 block Loftsgordon Ave.	PIO Stephanie Fryer	23:43:00
03/01/2024	Fight (In Progress)	2024-00086656	2222 E. Washington Ave.	PIO Stephanie Fryer	12:30:00
03/01/2024	Traffic incident/Road Rage	2024-86393	Tokay Blvd. at S. Whitney Way	PIO Stephanie Fryer	09:18:00
03/01/2024	Weapons Violation	2024-085825	200 block of State St.	PIO Stephanie Fryer	20:30:00
03/01/2024	Residential Burglary	2024-900620	600 block W. Main St.	PIO Stephanie Fryer	14:15:00
03/01/2024	Theft	2024-85843	State St. at Johnson St.	PIO Stephanie Fryer	20:15:00

sample police_reports.csv

Creating the React website

Integrates the Google Maps API, real-time weather data, and crime-mapping

- Features
  - Inclusive theme selection
  - Real-time weather updates
  - Routing
  - Crime awareness
- Development
  - Built with React for an interactive UI
  - Used google maps api for mapping/routing



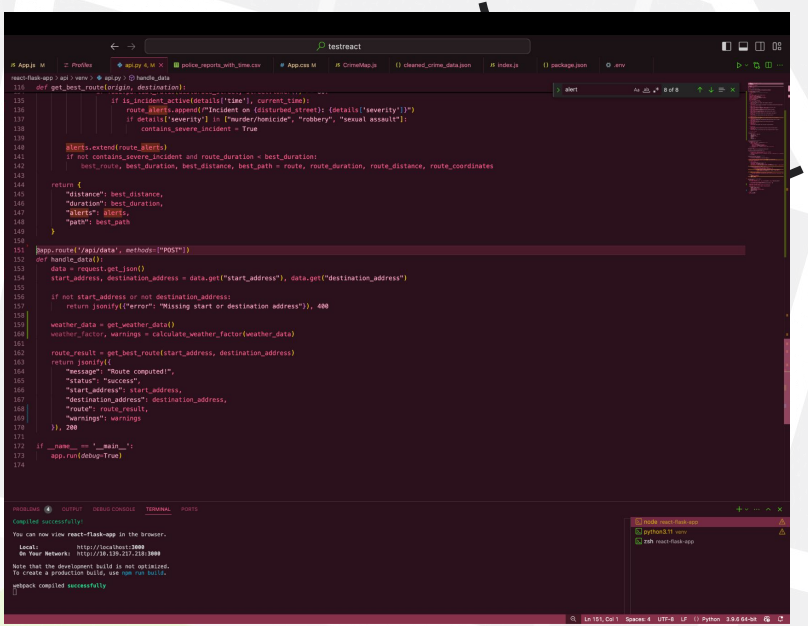
Linking it all with Flask

Set up Flask

- Create a Flask app with Flask and Flask-CORS to handle requests from React.
- Define API routes (@app.route) to handle GET and POST requests.
- Process data, such as user input or route calculations, and return JSON responses.
- Run the Flask app on a different port, e.g., 5000.

Connect React to Flask

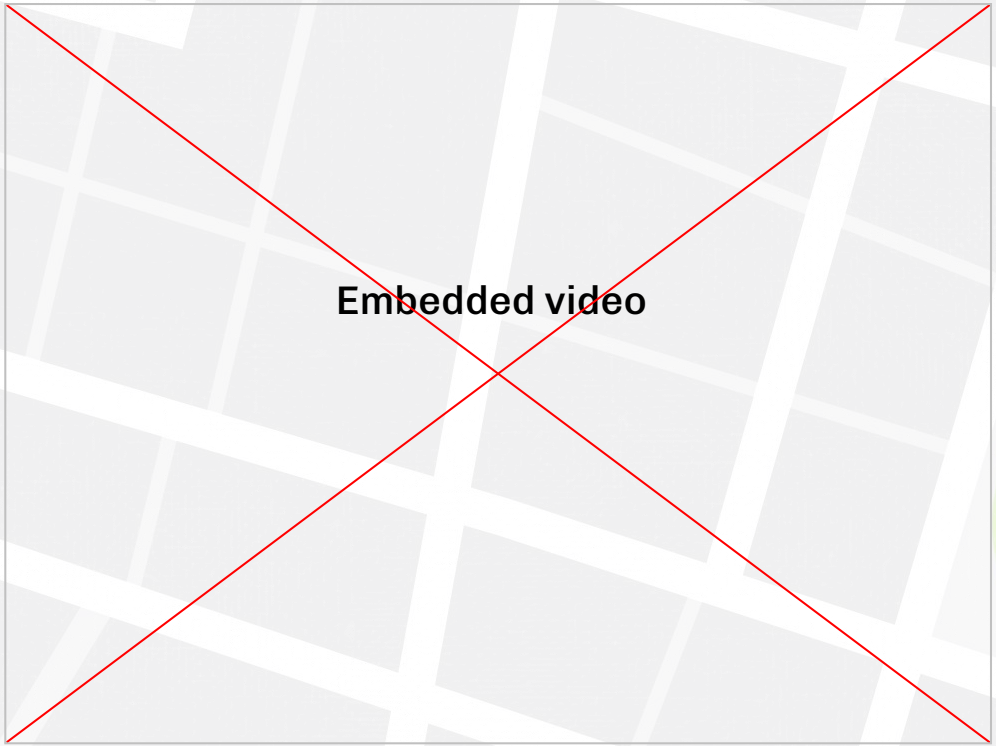
- Use fetch() or axios in React to send requests to Flask's API.
- Use useEffect to fetch data on component mount and useState to store responses.
- Send user inputs (e.g., locations) from React via POST requests to Flask.
- Display Flask's response data (e.g., route info, weather, crime reports) in React components.



03

Demonstration

SafeNav in action



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Takeaways + Future Ideas

SafeNav 2.0

- Add a dynamic element to the police reports data + have it update in real time
- Implement other modes of transportation (ex. cycling, driving)
- Incorporate filters
    - #1: No alert if low crime happened > 3 months ago, medium > 6 months, severe > 1 year ago
    - #2: Toggle between types of crime displayed
- Introduce a crowd-sourcing element: allow user input as to whether or not they are feeling unsafe and use ML predictive models to assess validity and use to guide further routes

THANK YOU!